COMPUTER NETWORK

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PES1UG24CS009

4A

ICMP

```
mininet> pingall
*** Ping: testing ping reachability
h1 -> h2 h3
h2 -> h1 h3
h3 -> h1 h2
*** Results: 0% dropped (6/6 received)
```

```
INFO:openflow.of_01:[00-00-00-00-00-01 2] connected
DEBUG:openflow.of_01:1 connection aborted
INFO:traffic_classification:Packet Classified: ICMP
INFO:traffic_classification:Stats -> ICMP:1 TCP:0 UDP:0 OTHER:0
INFO:traffic_classification:Traffic Distribution -> ICMP: 100.00% | TCP: 0.00% | UDP: 0.00%
INFO:traffic_classification:Dominant Traffic Type: ICMP
INFO:traffic_classification:Packet Classified: ICMP
INFO:traffic_classification:Stats -> ICMP:2 TCP:0 UDP:0 OTHER:0
INFO:traffic_classification:Traffic Distribution -> ICMP: 100.00% | TCP: 0.00% | UDP: 0.00%
INFO:traffic_classification:Dominant Traffic Type: ICMP
INFO:traffic_classification:Packet Classified: ICMP
INFO:traffic_classification:Stats -> ICMP:3 TCP:0 UDP:0 OTHER:0
INFO:traffic_classification:Traffic Distribution -> ICMP: 100.00% | TCP: 0.00% | UDP: 0.00%
INFO:traffic_classification:Dominant Traffic Type: ICMP
INFO:traffic_classification:Packet Classified: ICMP
INFO:traffic_classification:Stats -> ICMP:4 TCP:0 UDP:0 OTHER:0
INFO:traffic_classification:Traffic Distribution -> ICMP: 100.00% | TCP: 0.00% | UDP: 0.00%
INFO:traffic_classification:Dominant Traffic Type: ICMP
INFO:traffic_classification:Packet Classified: ICMP
INFO:traffic_classification:Stats -> ICMP:5 TCP:0 UDP:0 OTHER:0
INFO:traffic_classification:Traffic Distribution -> ICMP: 100.00% | TCP: 0.00% | UDP: 0.00%
INFO:traffic_classification:Dominant Traffic Type: ICMP
INFO:traffic_classification:Packet Classified: ICMP
INFO:traffic_classification:Stats -> ICMP:6 TCP:0 UDP:0 OTHER:0
INFO:traffic_classification:Traffic Distribution -> ICMP: 100.00% | TCP: 0.00% | UDP: 0.00%
INFO:traffic_classification:Dominant Traffic Type: ICMP
INFO:traffic_classification:Packet Classified: ICMP
INFO:traffic_classification:Stats -> ICMP:7 TCP:0 UDP:0 OTHER:0
INFO:traffic_classification:Traffic Distribution -> ICMP: 100.00% | TCP: 0.00% | UDP: 0.00%
INFO:traffic_classification:Dominant Traffic Type: ICMP
INFO:traffic_classification:Packet Classified: ICMP
```

Desktop 2

TCP

```
------------------------------------------------------------
Client connecting to 10.0.0.1, TCP port 5001
TCP window size: 85.3 KByte (default)
------------------------------------------------------------
[  1] local 10.0.0.2 port 37338 connected with 10.0.0.1 port 5001 (icwnd/mss/irtt=14/1448/15739)
[ ID] Interval       Transfer     Bandwidth
[  1] 0.0000-10.0161 sec  66.6 GBytes  57.1 Gbits/sec
mininet>
```

```
INFO:traffic_classification:Traffic Distribution -> ICMP: 100.00% | TCP: 0.00% | UDP: 0.00%
INFO:traffic_classification:Dominant Traffic Type: ICMP
INFO:traffic_classification:Packet Classified: TCP
INFO:traffic_classification:Stats -> ICMP:12 TCP:1 UDP:0 OTHER:0
INFO:traffic_classification:Traffic Distribution -> ICMP: 92.31% | TCP: 7.69% | UDP: 0.00%
INFO:traffic_classification:Dominant Traffic Type: ICMP
INFO:traffic_classification:Packet Classified: TCP
INFO:traffic_classification:Stats -> ICMP:12 TCP:2 UDP:0 OTHER:0
INFO:traffic_classification:Traffic Distribution -> ICMP: 85.71% | TCP: 14.29% | UDP: 0.00%
INFO:traffic_classification:Dominant Traffic Type: ICMP
```

UDP

```
mininet> h1 iperf -s -u &
------------------------------------------------------------
Server listening on TCP port 5001
TCP window size: 85.3 KByte (default)
------------------------------------------------------------
[  1] local 10.0.0.1 port 5001 connected with 10.0.0.2 port 37338 (icwnd/mss/irtt=14/1448/12647)
[ ID] Interval       Transfer     Bandwidth
[  1] 0.0000-10.0004 sec  66.6 GBytes  57.2 Gbits/sec
mininet> h2 iperf -c h1 -u
------------------------------------------------------------
Client connecting to 10.0.0.1, UDP port 5001
Sending 1470 byte datagrams, IPG target: 11215.21 us (kalman adjust)
UDP buffer size:  208 KByte (default)
------------------------------------------------------------
[  1] local 10.0.0.2 port 35869 connected with 10.0.0.1 port 5001
```

```
INFO:traffic_classification:Packet Classified: TCP
INFO:traffic_classification:Stats -> ICMP:12 TCP:2 UDP:0 OTHER:0
INFO:traffic_classification:Traffic Distribution -> ICMP: 85.71% | TCP: 14.29% | UDP: 0.00%
INFO:traffic_classification:Dominant Traffic Type: ICMP
INFO:traffic_classification:Packet Classified: UDP
INFO:traffic_classification:Stats -> ICMP:12 TCP:2 UDP:1 OTHER:0
INFO:traffic_classification:Traffic Distribution -> ICMP: 80.00% | TCP: 13.33% | UDP: 6.67%
INFO:traffic_classification:Dominant Traffic Type: ICMP
INFO:traffic_classification:Packet Classified: UDP
INFO:traffic_classification:Stats -> ICMP:12 TCP:2 UDP:2 OTHER:0
INFO:traffic_classification:Traffic Distribution -> ICMP: 75.00% | TCP: 12.50% | UDP: 12.50%
INFO:traffic_classification:Dominant Traffic Type: ICMP
```

Flow table

```
mininet> dpctl dump-flows
*** s1 ------------------------------------------------------------------------
 cookie=0x0, duration=6.732s, table=0, n_packets=1, n_bytes=42, idle_timeout=10, hard_timeout=30, priority=65535,arp,in_port
="s1-eth2",vlan_tci=0x0000,dl_src=56:11:45:0c:53:75,dl_dst=36:a3:85:3b:ca:1a,arp_spa=10.0.0.2,arp_tpa=10.0.0.1,arp_op=1 acti
ons=output:"s1-eth1"
 cookie=0x0, duration=6.685s, table=0, n_packets=1, n_bytes=42, idle_timeout=10, hard_timeout=30, priority=65535,arp,in_port
="s1-eth1",vlan_tci=0x0000,dl_src=36:a3:85:3b:ca:1a,dl_dst=56:11:45:0c:53:75,arp_spa=10.0.0.1,arp_tpa=10.0.0.2,arp_op=2 acti
ons=output:"s1-eth2"
 cookie=0x0, duration=6.477s, table=0, n_packets=1, n_bytes=42, idle_timeout=10, hard_timeout=30, priority=65535,arp,in_port
="s1-eth3",vlan_tci=0x0000,dl_src=7a:09:79:49:02:51,dl_dst=36:a3:85:3b:ca:1a,arp_spa=10.0.0.3,arp_tpa=10.0.0.1,arp_op=1 acti
ons=output:"s1-eth1"
 cookie=0x0, duration=6.433s, table=0, n_packets=1, n_bytes=42, idle_timeout=10, hard_timeout=30, priority=65535,arp,in_port
="s1-eth1",vlan_tci=0x0000,dl_src=36:a3:85:3b:ca:1a,dl_dst=7a:09:79:49:02:51,arp_spa=10.0.0.1,arp_tpa=10.0.0.3,arp_op=2 acti
ons=output:"s1-eth3"
 cookie=0x0, duration=6.220s, table=0, n_packets=1, n_bytes=42, idle_timeout=10, hard_timeout=30, priority=65535,arp,in_port
="s1-eth3",vlan_tci=0x0000,dl_src=7a:09:79:49:02:51,dl_dst=56:11:45:0c:53:75,arp_spa=10.0.0.3,arp_tpa=10.0.0.2,arp_op=1 acti
ons=output:"s1-eth2"
 cookie=0x0, duration=6.172s, table=0, n_packets=1, n_bytes=42, idle_timeout=10, hard_timeout=30, priority=65535,arp,in_port
="s1-eth2",vlan_tci=0x0000,dl_src=56:11:45:0c:53:75,dl_dst=7a:09:79:49:02:51,arp_spa=10.0.0.2,arp_tpa=10.0.0.3,arp_op=2 acti
ons=output:"s1-eth3"
 cookie=0x0, duration=6.775s, table=0, n_packets=0, n_bytes=0, idle_timeout=10, hard_timeout=30, arp,vlan_tci=0x0000,dl_src=
56:11:45:0c:53:75,dl_dst=36:a3:85:3b:ca:1a,arp_spa=10.0.0.2,arp_tpa=10.0.0.1,arp_op=1 actions=FLOOD
 cookie=0x0, duration=6.685s, table=0, n_packets=0, n_bytes=0, idle_timeout=10, hard_timeout=30, arp,vlan_tci=0x0000,dl_src=
36:a3:85:3b:ca:1a,dl_dst=56:11:45:0c:53:75,arp_spa=10.0.0.1,arp_tpa=10.0.0.2,arp_op=2 actions=FLOOD
 cookie=0x0, duration=6.523s, table=0, n_packets=0, n_bytes=0, idle_timeout=10, hard_timeout=30, arp,vlan_tci=0x0000,dl_src=
7a:09:79:49:02:51,dl_dst=36:a3:85:3b:ca:1a,arp_spa=10.0.0.3,arp_tpa=10.0.0.1,arp_op=1 actions=FLOOD
 cookie=0x0, duration=6.433s, table=0, n_packets=0, n_bytes=0, idle_timeout=10, hard_timeout=30, arp,vlan_tci=0x0000,dl_src=
36:a3:85:3b:ca:1a,dl_dst=7a:09:79:49:02:51,arp_spa=10.0.0.1,arp_tpa=10.0.0.3,arp_op=2 actions=FLOOD
 cookie=0x0, duration=6.266s, table=0, n_packets=0, n_bytes=0, idle_timeout=10, hard_timeout=30, arp,vlan_tci=0x0000,dl_src=
7a:09:79:49:02:51,dl_dst=56:11:45:0c:53:75,arp_spa=10.0.0.3,arp_tpa=10.0.0.2,arp_op=1 actions=FLOOD
 cookie=0x0, duration=6.172s, table=0, n_packets=0, n_bytes=0, idle_timeout=10, hard_timeout=3 Show hidden icons tci=0x0000,dl_src=
```